

RT Climatometer Project - Technical Documentation

BUT Networks and Telecommunications

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version 1.0.0 Python 3.10+ Framework Flask Database SQLite3

Introduction

The RT Climate Meter is a full-stack application designed to centralize, map, and analyze climate data from student living spaces. This project combines the development of dynamic interfaces, relational database management, and communication with third-party API services.

System Architecture

1. Technological Choices

- Backend: Python with the Flask micro-framework. Chosen for its flexibility and ability to handle asynchronous tasks (threading).
- Database: SQLite3. A robust, serverless engine, ideal for project portability between collaborators.
- Mapping: Leaflet.js. Use of Clustering plugins to manage the overlapping of markers in areas with high student density (e.g., university residences).
- Frontend: Custom design in CSS3 with the use of Glassmorphism (blur and transparency effects).

2. Data Model (Relational Schema)

The application relies on a standardized SQL structure to guarantee data integrity:

Table of Student Role

Residences,	Stores the identity (Last Name, First Name).
Weather	Stores the location (City, Address, Type) linked to a student.
Reports	Stores the climate history (Temperature, Humidity, Sky) associated with a residence.

Software Component Analysis

A. Weather Refresh Management (Threading)

To ensure a smooth user experience without network latency, the application uses a Thread daemon in the background:

- Process: When the server starts, an infinite loop is initiated. It queries the OpenWeatherMap API for each recorded city every 60 seconds.
- Optimization: Data is stored in a database. The user interface only consults the local database, thus avoiding unnecessary and costly API calls when loading pages.

B. Median Climate Algorithm

Rather than a simple average, we implemented median calculation to eliminate the influence of outliers (students on internships abroad or API sensor errors).

$$\tilde{x} = \begin{cases} x_{\lfloor \frac{n+1}{2} \rfloor} & \text{if } n \text{ is odd} \\ \frac{x_{\lfloor \frac{n}{2} \rfloor} + x_{\lfloor \frac{n}{2} \rfloor + 1}}{2} & \text{if } n \text{ is even} \end{cases}$$

C. Mapping & Geocoding

When adding a student, the application performs reverse geocoding via the API to retrieve the GPS coordinates (coordinates allow Leaflet to precisely position markers on the class map).

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Installation and Deployment

1. Cloning and the Environment

```
git clone [https://github.com/Raphoxrr28/Climatom-tre-RT.git] (https://github.com/Raphoxrr28/Climatom-tre-RT.git) cd
Climatom-tre-RT python -m venv venv # Activation (Windows) : venv\Scripts\activate # Activation (Linux/Mac) :
source venv/bin/activate
```

```
<style type="text/css">@media print {
  *, :after, :before {background: 0 0 !important;color: #000 !important;box-shadow: none !important;text-shadow: none !im
  a, a:visited {text-decoration: underline} a[href]:after {content: " (" attr(href) ")"}
}
```

```

abbr[title]:after {content: " (" attr(title) ")"}
a[href^="#"]:after, a[href^="javascript:"]:after {content: ""}
blockquote, pre {border: 1px solid #999;page-break-inside: avoid}
thead {display: table-header-group} img, tr {page-break-inside:
avoid} img {max-width: 100% !important} h2, h3, p {orphans:
3;widows: 3} h2, h3 {page-break-after: avoid}

} html {font-size: 12px} @media screen and (min-width: 32rem)
and (max-width: 48rem) {

    html {font-size: 15px}
} @media screen and (min-width: 48rem) {

    html {font-size: 16px}
} body {line-height: 1.85} .air-p, p {font-size: 1rem;margin-bottom: 1.3rem} .air-h1, .air-h2, .air-h3, .air-h4, h1,
h2, h3, h4 {margin: 1.414rem 0 .5rem;font-weight: inherit;line-height: 1.42} .air-h1, h1 {margin-top: 0;font-size:
3.998rem} .air-h2, h2 {font-size: 2.827rem} .air-h3, h3 {font-size: 1.999rem} .air-h4, h4 {font-size: 1.414rem} .air-
h5, h5 {font-size: 1.121rem} .air-h6, h6 {font-size: .88rem} .air-small, small {font-size: .707em} canvas, iframe, img,
select, svg, textarea, video {max-width: 100%} body {color: #444;font-family: 'Open Sans', Helvetica, sans-serif;font-
weight: 300;margin: 0;text-align: center} img {border-radius: 50%;height: 200px;margin: 0 auto;width: 200px} a,
a:visited {color: #3498db} a:active, a:focus, a:hover {color: #2980b9} pre {background-color: #fafafa;padding:
1rem;text-align: left} blockquote {margin: 0;border-left: 5px solid #7a7a7a;font-style: italic;padding: 1.33em;text-
align: left} li, ol, ul {text-align: left} p {color: #777}</style>

```